

Capture the Flag 3: Data-Driven UI

Flag 0 — Create Your Feature Branch

```
WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (main)
$ git checkout -b feature/data-ui
Switched to a new branch 'feature/data-ui'

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git branch
* feature/data-ui
  feature/widgets
  main
```

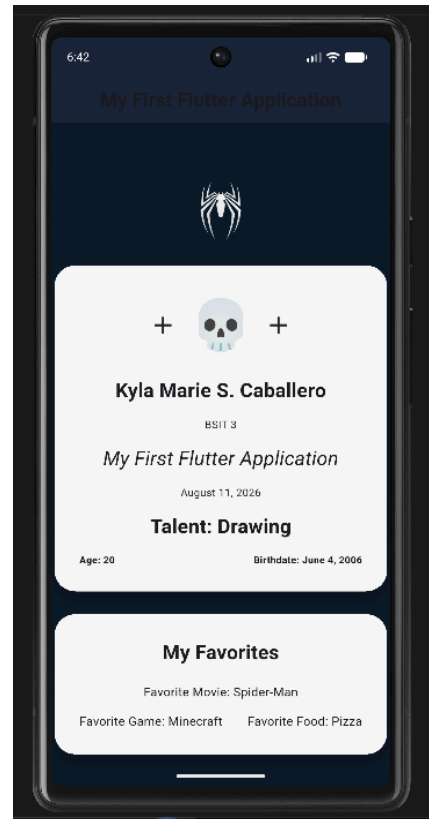
Flag 1 — Identify the Data

Kyla Marie S. Caballero	string
BSIT 3	string
My First Flutter...	string
August 11, 2026	string
20	int
June 4, 2006	string
Drawing	string
Spider-Man	string
Minecraft	string
Pizza	string
True	bool
1.60	double

Flag 2 — Create Your Data

```
const String name = "Kyla Marie S. Caballero";
const String course = "BSIT 3";
const int age = 20;
const String hobby = "Drawing";
const double height = 1.60;
const bool isStudent = true;

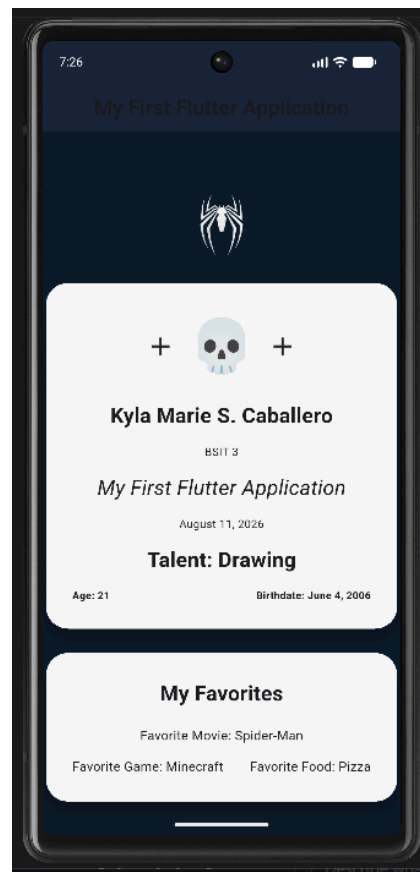
class MyApp extends StatelessWidget {
```



Flag 3 — Connect Data to UI

```
7 const String name = "Kyla Marie S. Caballero";
8 const String course = "BSIT 3";
9 const int age = 21;
10 const String hobby = "Drawing";
11 const double height = 1.60;
12 const bool isStudent = true;
```

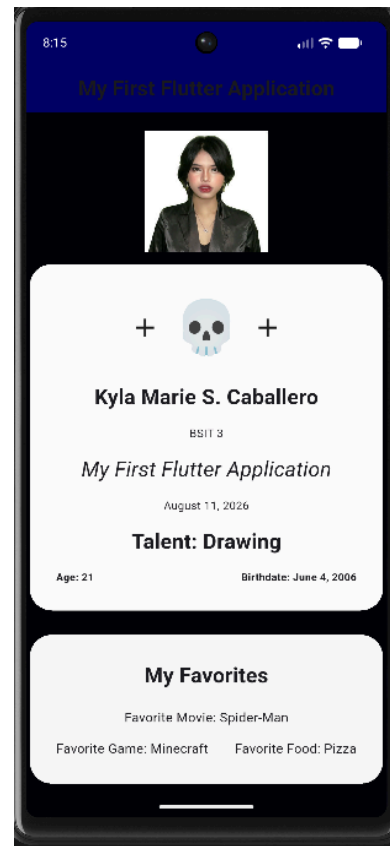
```
    "Age: $age",
    style: const TextStyle(
      fontSize: 12,
      fontWeight: FontWeight.bold,
    ), // TextStyle
  ), // Text
```



Flag 4 — Make the Image Data-Driven

```
const String name = "Kyla Marie S. Caballero";
const String course = "BSIT 3";
const int age = 21;
const String hobby = "Drawing";
const double height = 1.60;
const bool isStudent = true;
const String profileImage = "assets/kylamarie.jpg";
```

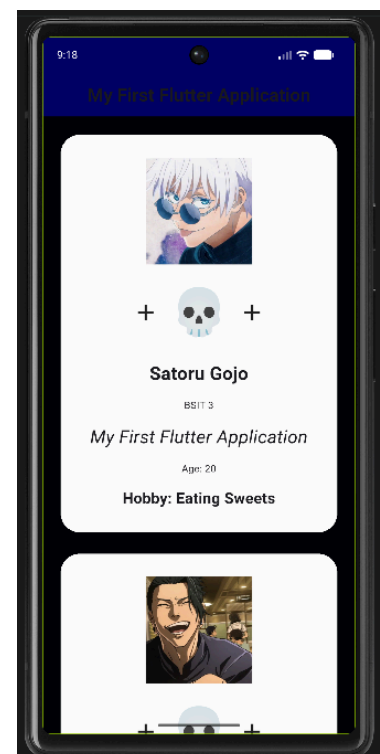
```
Padding(
  padding: const EdgeInsets.only(top: 20),
  child: Image.asset(
    profileImage,
    width: 140,
    height: 140,
  ), // Image.asset
), // Padding
```



Flag 5 — Five Profiles, One Design

```
const List<Profile> profiles = [
  Profile(
    image: "assets/satorugojo.jpg",
    name: "Satoru Gojo",
    course: "BSIT 3",
    age: 20,
    hobby: "Eating Sweets",
  ),
  Profile(
    image: "assets/sugurugeto.jpg",
    name: "Suguru Geto",
    course: "BSIT 3",
    age: 19,
    hobby: "Martial Arts",
  ),
  Profile(
    image: "assets/chosokamo.jpg",
    name: "Choso Kamo",
    course: "ME 4",
    age: 25,
    hobby: "Spending time with brothers",
  ),
];
```

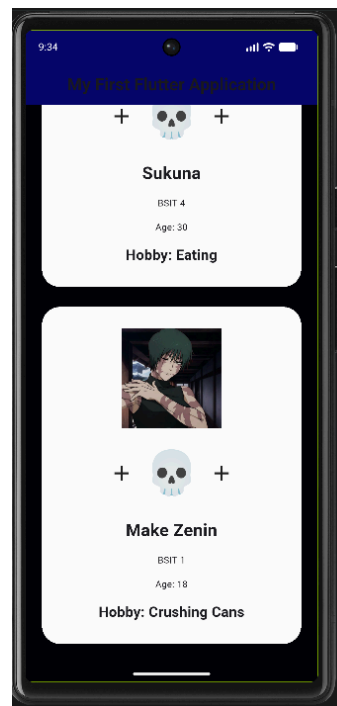
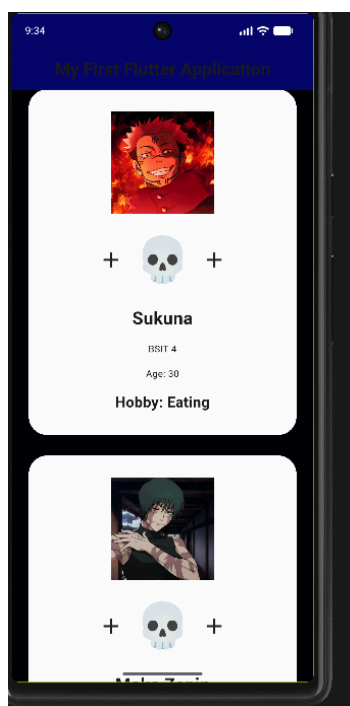
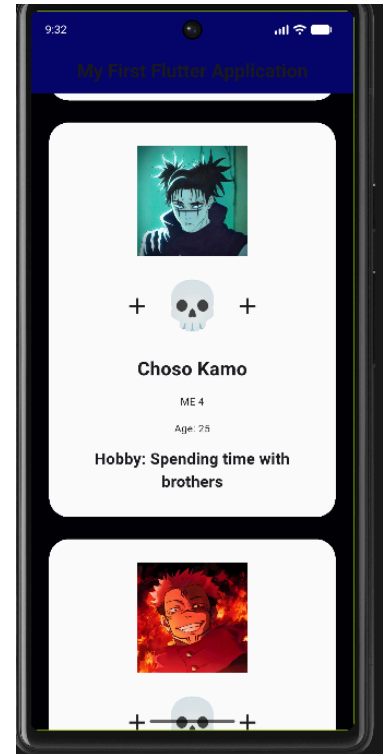
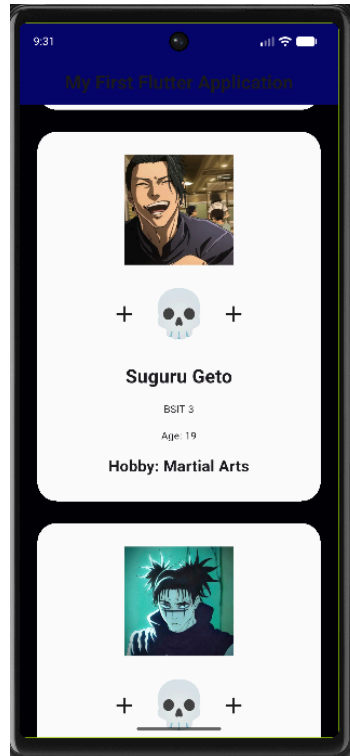
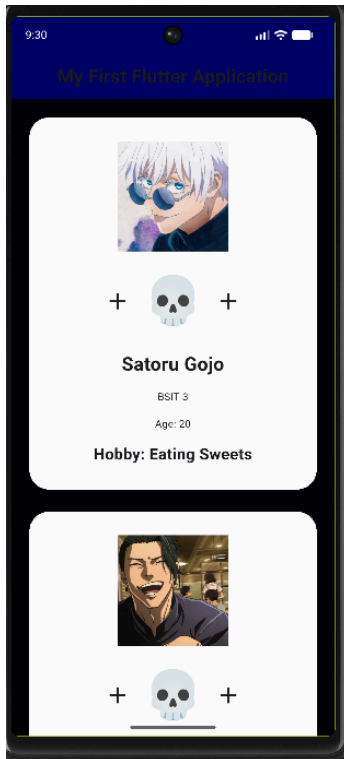
```
Profile(
  image: "assets/sukuna.jpg",
  name: "Sukuna",
  course: "BSIT 4",
  age: 30,
  hobby: "Eating",
),
Profile(
  image: "assets/makizinin.jpg",
  name: "Make Zenin",
  course: "BSIT 1",
  age: 18,
  hobby: "Crushing Cans",
),
];
```



Flag 6 — THE HIDDEN PROBLEM

```
// through all these profile cards
body: SingleChildScrollView(
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.center,
    children: [
      const SizedBox(height: 20),

      // Create one card for every profile
      for (final profile in profiles)
        Padding(
```

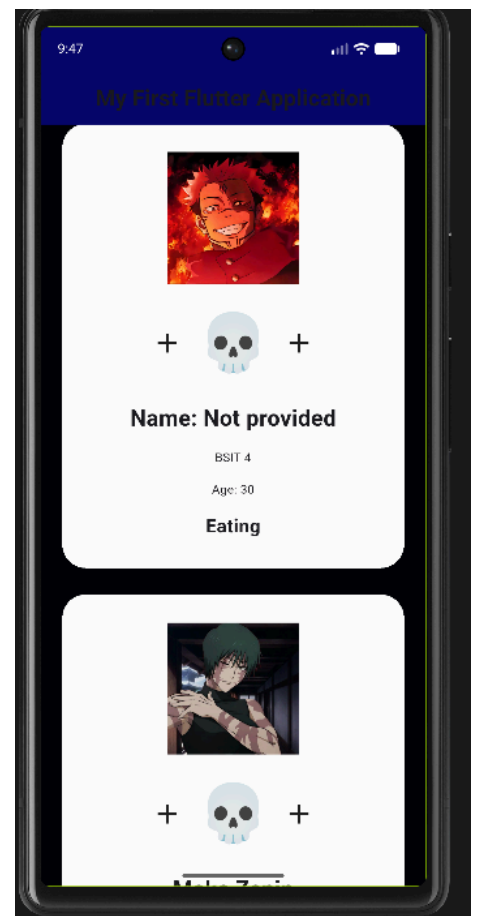
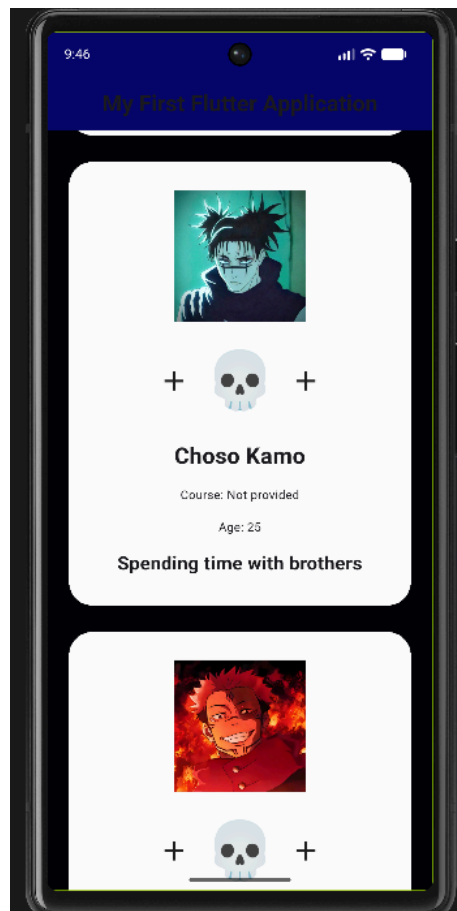
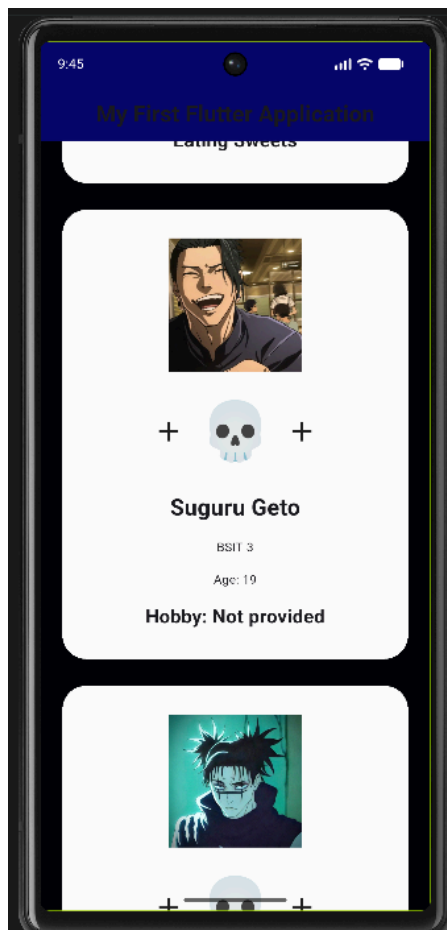


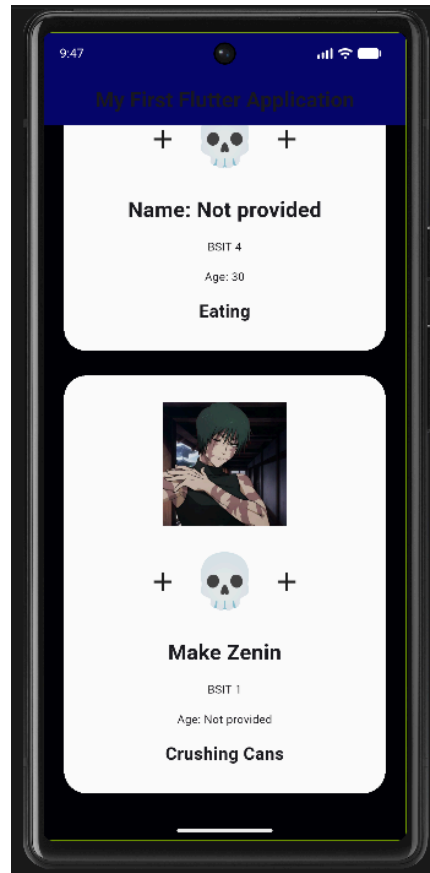
Flag 7 — Survive the Database

```
class Profile {  
  final String image;  
  final String? name;  
  final String? course;  
  final int? age;  
  final String? hobby;  
}
```

```
// Profile 2 - Hobby is missing  
Profile(  
  image: "assets/sugurugeto.jpg",  
  name: "Suguru Geto",  
  course: "BSIT 3",  
  age: 19,  
  hobby: null,  
),  
  
// Profile 3 - Course is missing  
Profile(  
  image: "assets/chosokamo.jpg",  
  name: "Choso Kamo",  
  course: null,  
  age: 25,  
  hobby: "Spending time with brothers",  
),
```

```
Profile(  
  image: "assets/sukuna.jpg",  
  name: null,  
  course: "BSIT 4",  
  age: 30,  
  hobby: "Eating",  
),  
  
// Profile 5 - Age is missing  
Profile(  
  image: "assets/makizenin.jpg",  
  name: "Make Zenin",  
  course: "BSIT 1",  
  age: null,  
  hobby: "Crushing Cans",  
),  
];
```





Flag 8 — Save Your Progress

```
MINGW64/c/IT314AB_Caballero

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ git branch
  feature/data-ui
* feature/widgets
  main

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ ^[[200~
bash: $'\E[200~': command not found

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ git switch feature/data-ui
error: Your local changes to the following files would be overwritten by checkout:
    FlutterActivities/my_first_flutter_app/lib/main.dart
Please commit your changes or stash them before you switch branches.
Aborting

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ AC

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ git status
on branch feature/widgets
your branch is up to date with 'origin/feature/widgets'.

changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   FlutterActivities/my_first_flutter_app/lib/main.dart
    modified:   FlutterActivities/my_first_flutter_app/pubspec.yaml
    modified:   FlutterActivities/my_first_flutter_app/test/widget_test.dart

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    FlutterActivities/my_first_flutter_app/assets/chosokamo.jpg
    FlutterActivities/my_first_flutter_app/assets/kylamarie.jpg
    FlutterActivities/my_first_flutter_app/assets/makizenin.jpg
    FlutterActivities/my_first_flutter_app/assets/satorugojo.jpg
    FlutterActivities/my_first_flutter_app/assets/sugurugeto.jpg
    FlutterActivities/my_first_flutter_app/assets/sukuna.jpg

no changes added to commit (use "git add" and/or "git commit -a")

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ git stash --help

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ git add .

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ git stash save
saved working directory and index state on feature/widgets: .

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ git status
on branch feature/widgets
your branch is up to date with 'origin/feature/widgets'.

nothing to commit, working tree clean
```

```
MINGW64/c/IT314AB_Caballero

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git status
on branch feature/data-ui
your branch is up to date with 'origin/feature/data-ui'.

changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   FlutterActivities/my_first_flutter_app/assets/chosokamo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/kylamarie.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/makizenin.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/satorugojo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sugurugeto.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sukuna.jpg
    modified:   FlutterActivities/my_first_flutter_app/pubspec.yaml
    modified:   FlutterActivities/my_first_flutter_app/test/widget_test.dart

Unmerged paths:
  (use "git restore --staged <file>..." to unstage)
  (use "git add <file>..." to mark resolution)
    both modified:   FlutterActivities/my_first_flutter_app/lib/main.dart

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git add .

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git status
on branch feature/data-ui
your branch is up to date with 'origin/feature/data-ui'.

changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   FlutterActivities/my_first_flutter_app/assets/chosokamo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/kylamarie.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/makizenin.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/satorugojo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sugurugeto.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sukuna.jpg
    modified:   FlutterActivities/my_first_flutter_app/lib/main.dart
    modified:   FlutterActivities/my_first_flutter_app/pubspec.yaml
    modified:   FlutterActivities/my_first_flutter_app/test/widget_test.dart

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git commit -m "Feat: Added Flutter Project Act 2 - Variables"
[feature/data-ui 56ee11] Feat: Added Flutter Project Act 2 - Variables
 9 files changed, 417 insertions(+), 30 deletions(-)
 create mode 100644 FlutterActivities/my_first_flutter_app/assets/chosokamo.jpg
 create mode 100644 FlutterActivities/my_first_flutter_app/assets/kylamarie.jpg
 create mode 100644 FlutterActivities/my_first_flutter_app/assets/makizenin.jpg
 create mode 100644 FlutterActivities/my_first_flutter_app/assets/satorugojo.jpg
 create mode 100644 FlutterActivities/my_first_flutter_app/assets/sugurugeto.jpg
 create mode 100644 FlutterActivities/my_first_flutter_app/assets/sukuna.jpg

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git push origin feature/data-ui
Enumerating objects: 25, done.
Counting objects: 100% (25/25), done.
Delta compression using up to 8 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (16/16), 336.96 KiB | 12.48 MiB/s, done.
Total 16 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 3 local objects.
To https://github.com/mykyliecaballero/IT314AB_Caballero.git
   8cb92..56ee11  feature/data-ui -> feature/data-ui
```

```
MINGW64/c/IT314AB_Caballero

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/widgets)
$ git checkout feature/data-ui
Switched to branch 'feature/data-ui'
Your branch is up to date with 'origin/feature/data-ui'.

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git stash apply
Auto-merging FlutterActivities/my_first_flutter_app/lib/main.dart
CONFLICT (content): Merge conflict in FlutterActivities/my_first_flutter_app/lib/main.dart
On branch feature/data-ui
Your branch is up to date with 'origin/feature/data-ui'.

changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   FlutterActivities/my_first_flutter_app/assets/chosokamo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/kylamarie.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/makizenin.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/satorugojo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sugurugeto.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sukuna.jpg
    modified:   FlutterActivities/my_first_flutter_app/pubspec.yaml
    modified:   FlutterActivities/my_first_flutter_app/test/widget_test.dart

Unmerged paths:
  (use "git restore --staged <file>..." to unstage)
  (use "git add <file>..." to mark resolution)
    both modified:   FlutterActivities/my_first_flutter_app/lib/main.dart

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git status
on branch feature/data-ui
your branch is up to date with 'origin/feature/data-ui'.

changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   FlutterActivities/my_first_flutter_app/assets/chosokamo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/kylamarie.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/makizenin.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/satorugojo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sugurugeto.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sukuna.jpg
    modified:   FlutterActivities/my_first_flutter_app/pubspec.yaml
    modified:   FlutterActivities/my_first_flutter_app/test/widget_test.dart

Unmerged paths:
  (use "git restore --staged <file>..." to unstage)
  (use "git add <file>..." to mark resolution)
    both modified:   FlutterActivities/my_first_flutter_app/lib/main.dart

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git status
on branch feature/data-ui
your branch is up to date with 'origin/feature/data-ui'.

changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   FlutterActivities/my_first_flutter_app/assets/chosokamo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/kylamarie.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/makizenin.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/satorugojo.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sugurugeto.jpg
    new file:   FlutterActivities/my_first_flutter_app/assets/sukuna.jpg
```